

INTERNSHIP SUMMARY REPORT

on the activities carried out during the internship

Host organisation S.A. "Orange Systems" _____
Faculty of *Computers, Informatics, and Microelectronics* _____
Study programme *Software Engineering* _____
Mode of study *full time* _____
Year of study: 3 _____ **Group:** FAF-243 _____
Internship period: from 01.09.2026 _____ to 26.09.2026 _____
Activities carried out and results obtained:

Activities carried out:

Over four weeks in September 2026, our team of five designed and built SecretShare, a web app for sending a credential to one named colleague so it can be read exactly once. Within the team I was responsible for security. I set up the initial project structure, Docker setup and the first version of the message relay. I then implemented the API hardening: rate limiting, central error handling, a 64 KB request size limit, security response headers and generic authentication errors. I built the PostgreSQL layer and an append-only audit log, added a Content Security Policy with per-request nonces to the Next.js frontend, and extended the CI pipeline to run security tests against real PostgreSQL and Redis. In the final week I wrote the security documentation.

Results obtained:

The team delivered a working MVP in which the server never sees plaintext and releases a secret only to the intended recipient, after which it is destroyed. My contribution includes 55 automated API security tests (single delivery, identifier entropy, size limit, expiry, audit logging, access logs, error disclosure) and 11 CSP tests for the frontend. I also produced the security requirements (28), STRIDE threat model, risk register (23 risks), security controls catalogue (20 API controls plus web controls), incident response plan, OWASP checklist and cryptographic specification, and aligned the project report with the implemented system.

Difficulties encountered during the internship:

The most serious issue I found was that the secret's identifier travelled in the URL and ended up in the web server access log; I moved it out of the URL, added a log filter and tests, and recorded the evidence. Authentication errors revealed why a token was rejected, and the audit setting was ignored by the secret routes; both were fixed and covered by tests. A dependency was incompatible with Python 3.14, and the security tests needed real databases in CI before they were meaningful. The documentation also drifted from the code as the design changed (Keycloak and Traefik replaced the original plan), so I had to re-check every claim against the implementation.

Conclusions and recommendations:

The internship taught me that security has to be designed in from the first day and proven with tests and evidence, not only described. Writing the threat model and risk register after implementation showed me that documentation is most useful when written alongside the code. The main drawback was the short duration, which left the known limitations (extractable device keys, no secret scanning in CI, no multi-factor authentication) as planned work for the PBL semester. I suggest a slightly longer programme or starting the infrastructure and threat modelling before the first week.

The company provided a supportive environment for reaching the internship's goals. Our mentor was helpful, showed real interest in the project, and gave feedback that improved both the implementation and the security analysis.

Date of completion: 26 / 09 / 2026

Signature of the student: Kushnirenko Ekaterina